

Yıldız Technical University
Computer Engineering Department
2023-2024 Spring
BLM3620 Digital Signal Processing
Homework 4, Form: A

Name: _____

Surname: _____

Student I.D.: _____

Signature: _____

Signal	FT
$x(t)$	$X(w)$
$y(t)$	$Y(w)$
$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(w)e^{jw t} dw$	$\int_{-\infty}^{\infty} x(t)e^{-jw t} dt$
$\delta(t)$	1
$\Pi(t) = \begin{cases} 0, & t > \frac{1}{2} \\ 1, & t \leq \frac{1}{2} \end{cases}$	$\text{sinc}\left(\frac{w}{2\pi}\right)$
$\Lambda(t) = \begin{cases} 0, & t > 1 \\ 1 - t , & t \leq 1 \end{cases}$	$\text{sinc}^2\left(\frac{w}{2\pi}\right)$
1	$2\pi\delta(w)$
$e^{-jw_0 t}$	$2\pi\delta(w - w_0)$
$e^{- a t}u(t)$	$\frac{1}{ a + jw}$
$e^{ a t}u(-t)$	$\frac{1}{ a - jw}$
$e^{- at }$	$\frac{2 a }{ a ^2 + w^2}$
$e^{-\pi t^2}$	$e^{-\frac{w^2}{4\pi}}$
$u(t)$	$\pi\delta(w) + \frac{1}{jw}$
$\cos(w_0 t)$	$\pi\delta(w + w_0) + \pi\delta(w - w_0)$
$\sin(w_0 t)$	$j\pi\delta(w + w_0) - j\pi\delta(w - w_0)$
$\sum_{n=-\infty}^{\infty} \delta(t - n)$	$\sum_{n=-\infty}^{\infty} \delta(w - n)$
$t^n x(t)$	$j^n \frac{d^n X(w)}{dw^n}$
$ t $	$-\frac{1}{w^2}$
$\frac{1}{1+t^2}$	$\pi e^{- w }$
$ax(t) + by(t)$	$aX(w) + bY(w)$
$x(t - a)$	$X(w)e^{-jwa}$
$x\left(\frac{t}{a}\right)$	$ a X(aw)$
$x(t) * y(t)$	$X(w)Y(w)$
$x(t)y(t)$	$\frac{1}{2\pi} X(w) * Y(w)$
$x(t)e^{jw_0 t}$	$X(w - w_0)$
$x^*(t)$	$X^*(-w)$
$x(-t)$	$X(-w)$
$\frac{dx(t)}{dt}$	$jwX(w)$
$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{jw} X(w)$
$x(t)$ is real	$X(w) = X^*(-w)$
$x(t)$ is real and even	$X(w)$ is real and even
$x(t)$ is real and odd	$X(w)$ is imaginary and odd
$\int_{-\infty}^{\infty} x(t) ^2 dt$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(w) ^2 dw$

Sequence	DTFT
$x[n]$	$X(e^{jw})$
$y[n]$	$Y(e^{jw})$
$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{jw}) e^{jwn} dw$	$\sum_{n=-\infty}^{\infty} x[n] e^{-jwn}$
$\delta[n]$	1
$\delta[n-a]$	e^{-jwa}
$\sum_{m=-\infty}^{\infty} \delta[n-m]$	$2\pi \sum_{k=-\infty}^{\infty} \delta(w-2\pi k)$
e^{-jan}	$2\pi \sum_{k=-\infty}^{\infty} \delta(w+a-2\pi k)$
$u[n]$	$\frac{1}{1-e^{-jw}} + \pi \sum_{k=-\infty}^{\infty} \delta(w-2\pi k)$
$a^n u[n], a < 1$	$\frac{1}{1-ae^{-jw}}$
$-a^n u[-n-1], a > 1$	$\frac{1}{1-ae^{-jw}}$
$(n+1)a^n u[n], a < 1$	$\frac{1}{(1-ae^{-jw})^2}$
$\cos(na)$	$\pi \sum_{k=-\infty}^{\infty} \delta(w-a-2\pi k) + \delta(w+a-2\pi k)$
$\sin(na)$	$\frac{\pi}{j} \sum_{k=-\infty}^{\infty} \delta(w-a-2\pi k) - \delta(w+a-2\pi k)$
$x[n] = \begin{cases} 0, & n > N \\ 1, & n \leq N \end{cases}$	$\frac{\sin(w(N+\frac{1}{2}))}{\sin(\frac{w}{2})}$
$\frac{\sin(An)}{\pi n}$	$X(e^{jw}) = \begin{cases} 1, & w \leq A \\ 0, & A < w \leq \pi \end{cases}, X(e^{jw}) \text{ periodic by } 2\pi$
$ax[n] + by[n]$	$aX(e^{jw}) + bY(e^{jw})$
$x[n-n_0]$	$X(e^{jw}) e^{-jwn_0}$
$x[n] e^{jwn_0}$	$X(e^{j(w-w_0)})$
$x^*[n]$	$X^*(e^{-jw})$
$x[-n]$	$X(e^{-jw})$
$x[n] * y[n]$	$X(e^{jw}) Y(e^{jw})$
$x[n] y[n]$	$\frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j(w-\theta)}) d\theta$
$x[n] - x[n-1]$	$(1-e^{-jw}) X(e^{jw})$
$\sum_{k=-\infty}^{\infty} x[k]$	$\frac{1}{(1-e^{-jw})} + \pi X(1) \sum_{k=-\infty}^{\infty} \delta(w-2\pi k)$
$nx[n]$	$j \frac{dX(e^{jw})}{dw}$
$x[n]$ is real	$X(e^{jw}) = X^*(e^{-jw})$
$x[n]$ is real and even	$X(e^{jw})$ is real and even
$x[n]$ is real and odd	$X(e^{jw})$ is imaginary and odd
$\sum_{n=-\infty}^{\infty} x[n] ^2$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} X(e^{jw}) ^2 dw$

Sequence	DFT
$x[n]$	$X[k]$
$y[n]$	$Y[k]$
$\frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi \frac{kn}{N}}$	$\sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}}$
$\sum_{k=-\infty}^{\infty} \delta[n + Nk]$	1 (period N)
1 (period N)	$N \sum_{m=-\infty}^{\infty} \delta[k + Nm]$
$e^{j2\pi k_0 n}$	$N \delta[((k - k_0))_N]$
$\cos\left(2\pi \frac{k_0 n}{N}\right)$	$\frac{N}{2} (\delta[((k - k_0))_N] + \delta[((k + k_0))_N])$
$ax[n] + by[n]$	$aX[k] + bY[k]$
$x[((n - m))_N]$	$X[k] e^{-j2\pi \frac{km}{N}}$
$X[n]$	$NX[((-k))_N]$
$x[n]y[n]$	$\frac{1}{N} X[k] \otimes Y[k]$
$x[n] \otimes y[n]$	$X[k]Y[k]$
$x^*[n]$	$X^*[((-k))_N]$
$x[((-n))_N]$	$X^*[k]$
$\text{Re}\{x[n]\}$	$\frac{1}{2} (X[((k))_N] + X^*[((-k))_N])$
$j \text{Im}\{x[n]\}$	$\frac{j}{2} (X[((k))_N] - X^*[((-k))_N])$
$\frac{1}{2} (x[((n))_N] + x^*[((-n))_N])$	$\text{Re}\{X[k]\}$
$\frac{j}{2} (x[((n))_N] - x^*[((-n))_N])$	$j \text{Im}\{X[k]\}$
$\sum_{n=0}^{N-1} x[n] ^2$	$\frac{1}{N} \sum_{k=0}^{N-1} X[k] ^2$

Sequence	ZT	Region of Convergence
$x[n]$	$X(z)$	
$y[n]$	$Y(z)$	
$a^n u[n]$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
$-na^n u[-n - 1]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
$\delta[n]$	1	All z
$\delta[n - n_0]$	z^{-n_0}	All z
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$\cos(\omega_0 n) u[n]$	$\frac{1 - z^{-1} \cos(\omega_0)}{1 - 2z^{-1} \cos(\omega_0) + z^{-2}}$	$ z > 1$
$\sin(\omega_0 n) u[n]$	$\frac{z^{-1} \sin(\omega_0)}{1 - 2z^{-1} \cos(\omega_0) + z^{-2}}$	$ z > 1$
$a^n \cos(\omega_0 n) u[n]$	$\frac{1 - az^{-1} \cos(\omega_0)}{1 - 2az^{-1} \cos(\omega_0) + a^2 z^{-2}}$	$ z > a $
$a^n \sin(\omega_0 n) u[n]$	$\frac{az^{-1} \sin(\omega_0)}{1 - 2az^{-1} \cos(\omega_0) + a^2 z^{-2}}$	$ z > a $
$Ax[n] + By[n]$	$AX(z) + BY(z)$	
$x[n - n_0]$	$X(z)z^{-n_0}$	
$a^n x[n]$	$X(a^{-1}z)$	
$x^*[n]$	$X^*(z^*)$	
$x[-n]$	$X(z^{-1})$	
$x[n] * y[n]$	$X(z)Y(z)$	
$nx[n]$	$-z \frac{dX(z)}{dz}$	
$x[n]$ is causal	$x(0) = \lim_{z \rightarrow \infty} X(z)$	
$x[n]$ is causal	$x(\infty) = \lim_{z \rightarrow 1} [z - 1]X(z)$	

1. Determine the fundamental period of $x[n]$.

$$x[n] = \cos\left(\frac{n\pi}{10}\right) + \sin\left(\frac{n\pi}{15}\right)$$

2. Determine whether or not the signal $x[n]$ is periodic.

$$x[n] = \sin(\sqrt{2} + 0.2n)$$

3. Given that real valued signal $x_1[n]$ is even by definition $x_1[n] = x_1[-n]$, and real valued signal $x_2[n]$ is odd by definition $x_2[n] = -x_2[-n]$, determine symmetry (even/odd) of $y[n]$.

$$y[n] = x_1[n] \cdot x_2[n]$$

4. Given that the power of real valued signal $x[n]$ is defined as $P = \sum_{n=-\infty}^{\infty} x^2[n]$, compute the power in $y[n]$.

$$y[n] = 2^n \cdot u[-n]$$

5. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is time (shift)-invariant.

$$y[n] = x[n] \cdot u[n]$$

6. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is linear.

$$y[n] = \text{Im}(x[n])$$

7. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is casual.

$$y[n] = x[|n|]$$

8. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine unit sample response $h[n]$ of the system.

$$y[n] = 0.5y[n-1] + 4x[n-2]$$

9. The responses of a linear time (shift)-invariant system to specified inputs are defined as follows:

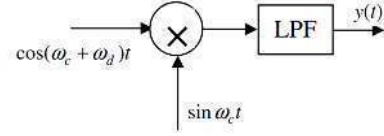
input		response
name	symbol	
unit sample	$\delta[n]$	$h[n]$
unit step	$u[n]$	$s[n]$

calculate $h[n]$ for the system, given that $s[n] = u[n] - u[n-5]$.

10. Find the Fourier transform of $x(t)$.

$$x(t) = \begin{cases} \frac{1}{2}, & -T < t < T \\ 0, & \text{other} \end{cases}$$

11. Given that the cut-off frequency of the low pass filter (LPF) is w_c evaluate the output $y(t)$. Note: LPF allows frequency values between $-w_c < w < w_c$.



$$x[n] = 2\delta[n+2] - \delta[n+1] + 3\delta[n] - \delta[n-1] + 2\delta[n-2]$$

12. Given that $x[n] \xleftrightarrow{DTFT} X(e^{jw})$ is a DTFT pair, evaluate $X(e^{jw})|_{w=\pi}$ without explicitly finding $X(e^{jw})$.

$$x[n] = 2\delta[n+2] - \delta[n+1] + 3\delta[n] - \delta[n-1] + 2\delta[n-2]$$

13. Given that $x[n] \xleftrightarrow{DTFT} X(e^{jw})$ and $y[n] \xleftrightarrow{DTFT} Y(e^{jw})$ are DTFT pairs, prove the convolution theorem.

$$x[n] * y[n] \xleftrightarrow{DTFT} X(e^{jw})Y(e^{jw})$$

14. Find the inverse DTFT of $X(e^{jw})$.

$$X(e^{jw}) = \cos^2(w)$$

15. Given the 6-point sequence $x[n] = [4, -1, 4, -1, 4, -1]$, determine its 6-point DFT sequence $X[k]$.

16. If the 4-point DFT an unknown length-4 sequence $v[n]$ is $V[k] = [1, 4 + j, -1, 4 - j]$, determine $v[n]$.

17. Find z-transforms of $x[n]$.

$$x[n] = 6\delta[n] - 7\delta[n-3] - 2\delta[n] - 9\delta[n-5]$$

18. If the region of convergence (ROC) for any $x[n] \xleftrightarrow{ZT} X(z)$ z-transform pair includes the unit circle in the complex plane then, $x[n] \xleftrightarrow{DTFT} X(e^{jw})$ DTFT pair can also be calculated (converges). Given that the following $X(z)$ includes the unit circle in its region of convergence, evaluate DTFT of $x[n]$ at $w = \pi$.

$$X(z) = \frac{z + 2z^{-2} + z^{-3}}{1 - 3z^{-4} + z^{-5}}$$

19. Evaluate $h[n] * x[n]$ using the convolution property of z-transform.

$$h[n] = (0.5)^n u[n]$$

$$x[n] = 3^n u[-n]$$

20. Given that $x[n] \xleftrightarrow{ZT} X(z)$ is a z-transform pair find $x[n]$.

$$X(z) = 2 + 5(z^2 + z^{-2})$$

21. Given that $x[n] \xleftrightarrow{ZT} X(z)$ is a z-transform pair find $x[n]$ for $|z| > 2$.

$$X(z) = \frac{1}{1 + 3z^{-1} + 2z^{-2}}$$

22. A continuous-time sinusoid $a_1(t) = \cos(w_1 t + 0.1\pi)$ is sampled at $f_{s_1} = 40 \text{ Hz}$ to give $a_1[n]$, and a second continuous-time sinusoid $a_2(t) = \cos(w_2 t + 0.1\pi)$ is sampled at $f_{s_2} = 50 \text{ Hz}$ to give $a_2[n]$. If $w_2 = 30\pi \text{ rad/s}$, determine w_1 so that $a_1[n] = a_2[n]$. Assume there is no aliasing when sampling $a_1(t)$.

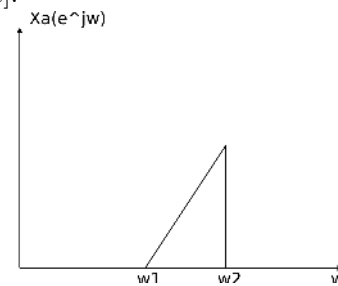
23. A complex bandpass filter is constructed by frequency shifting a running sum filter. evaluate and plot the magnitude frequency response of the complex bandpass filter $|H_B(e^{jw})|$.

$$h[n] = \sum_{k=0}^4 \delta[n - k]$$

$$h_B[n] = h[n]e^{jw_0 n}$$

$$h_B[n] \xleftrightarrow{DTFT} H_B(e^{jw})$$

24. A complex bandpass analog signal $x_a(t)$ has Fourier transform that is non-zero over the range of $[w_1, w_2]$. The signal is sampled to produce the sequence $x[n] = x_a(nT_s)$. What is the smallest sampling frequency that can be used so that $x_a(t)$ may be recovered from its samples $x[n]$.



25. Plot STFT (Short Time Fourier Transform) representation of a 1D chirp signal with different window sizes. Compare the results with FT of the same signal. Comment on what might be an optimal window size.

1. Determine the fundamental period of $x[n]$.

$$x[n] = \cos\left(\frac{n\pi}{10}\right) + \sin\left(\frac{n\pi}{15}\right)$$

$$x[n] \rightarrow \cos(n\omega) + \sin(n\omega)$$

$$\cos(n\omega) \rightarrow \cos\left(\frac{n\pi}{10}\right) \rightarrow \omega = \frac{\pi}{10}$$

$$\sin(n\omega) \rightarrow \sin\left(\frac{n\pi}{15}\right) \rightarrow \omega = \frac{\pi}{15}$$

$$\frac{2\pi}{T} = \omega \rightarrow T = \frac{2\pi}{\omega}$$

$$T_1 = \frac{2\pi}{\frac{\pi}{10}} \rightarrow 20$$

$$T_2 = \frac{2\pi}{\frac{\pi}{15}} \rightarrow 30$$

LCM(20, 30) $\rightarrow$ 60 fundamental period //

2. Determine whether or not the signal $x[n]$ is periodic.

$$x[n] = \sin(\sqrt{2} + 0.2n)$$

$$x[n] \rightarrow x[n+T]$$

$$\sqrt{2} + 0.2(n+T) \rightarrow \sqrt{2} + 0.2n$$

$$0.2T = 2\pi k$$

$$T = \frac{2\pi k}{0.2}$$

$$T = \frac{10\pi k}{\pi}$$

$$T = 10k$$

T 'nin 10'un tam katı olması gerekir
 $\sqrt{2}$ irrasyonel bir sayı olduğundan dolayı bu ifade
bir tam sayı sonucu vermez

3. Given that real valued signal $x_1[n]$ is even by definition $x_1[n] = x_1[-n]$, and real valued signal $x_2[n]$ is odd by definition $x_2[n] = -x_2[-n]$, determine symmetry (even/odd) of $y[n]$.

$$y[n] = x_1[n] \cdot x_2[n]$$

Signal çift ise $\rightarrow x[n] \rightarrow x_1[-n]$

Signal tek ise $\rightarrow -x[n] \rightarrow x_2[-n]$

$$y[-n] = x_1[-n] \cdot x_2[-n]$$

$$\begin{array}{cc} \downarrow & \downarrow \\ x_1[n] & -x_2[n] \end{array}$$

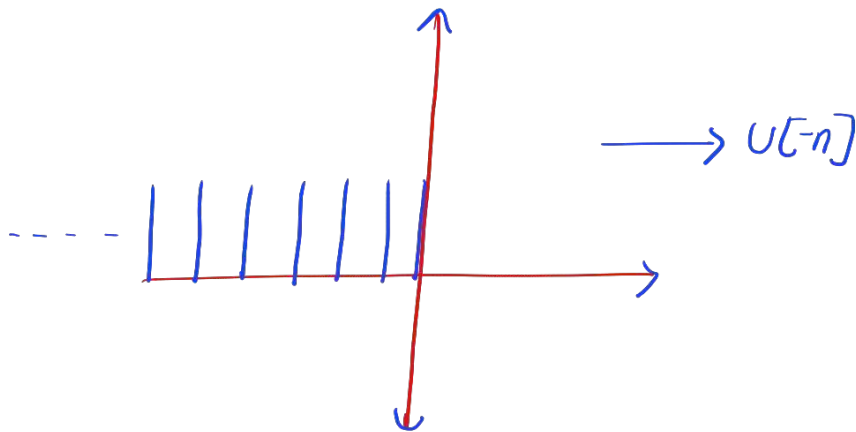
$$y[-n] = -(x_1[n] \cdot x_2[n])$$

$$y[n] = x_1[n] \cdot x_2[n]$$

$\rightarrow y[-n] = -y[n]$ olarak çıktı
 $\rightarrow$ bu yüzden tek
işaret

4. Given that the power of real valued signal $x[n]$ is defined as $P = \sum_{n=-\infty}^{\infty} x^2[n]$, compute the power in $y[n]$.

$$y[n] = 2^n \cdot u[-n]$$



$$x^2[n] \rightarrow (2^n)^2 \rightarrow 4^n$$

$$P \rightarrow \sum_{n=-\infty}^0 4^n \rightarrow \frac{1}{1-4^{-1}} = \frac{1}{1-\frac{1}{4}} = \frac{1}{3/4} \rightarrow 4/3 //$$

$$\sum_{n=-\infty}^0 r^n \rightarrow \frac{1}{1-r}$$

normalde bu formül $|r| < 1$ için geçerli
diğer durumlarda
çözümüze ıvırk olur
ama $n \leq 0$ olduğu
için böyle bir
dönüşüm yaptık

5. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is time (shift)-invariant.

$$y[n] = x[n] \cdot u[n]$$

$$y[n] \rightarrow x[n-n_0] \cdot u[n] \rightarrow a_1 \text{ signal}$$

$$y[n-n_0] \rightarrow x[n-n_0] \cdot u[n-n_0] \rightarrow a_2 \text{ signal}$$

a_1 ve a_2 birbirine eşit değil
bu yüzden zaman kaydırmalı
sistem değildir

6. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is linear.

$$y[n] = \text{Im}(x[n])$$

$$x_1[n] \rightarrow a_1 + j b_1$$

$$x_2[n] \rightarrow a_2 + j b_2$$

$$x_1[n] + x_2[n] = (a_1 + a_2) + j(b_1 + b_2)$$

$$\begin{cases} y_1[n] \rightarrow \text{Im}(x_1[n]) \rightarrow b_1 \\ y_2[n] \rightarrow \text{Im}(x_2[n]) \rightarrow b_2 \end{cases} \quad (+)$$

$$y[n] \rightarrow \text{Im}(x_1[n] + x_2[n]) \rightarrow b_1 + b_2$$

→ Bu ikisinin sonuçları
aynı çıktıyı için
linear bir sistemdir
deniz

7. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine whether or not the following systems is casual.

$$y[n] = x[[n]]$$

Nedensel olup olmadığına bakmak için sistemin mevcut ve geçmiş değerlere bağlı olup olmadığına bakılır

$n = -1$ için $x[1]$ gelecekteki bir değere bağlıdır

$n = 2$ için $x[2]$ mevcut bir değere bağlıdır

Sistem nedensel değildir

8. Given that $x[n]$ is the system input and $y[n]$ is the system output, determine unit sample response $h[n]$ of the system.

$$y[n] = 0.5y[n-1] + 4x[n-2]$$

$$x[n] = \delta[n]$$

$$x[0] = 1$$

$$x[n] = 0 \quad (n \neq 0)$$

$$n=0 \text{ i.e. } y[0] = 0.5y[-1] + 4x[-2] \rightarrow y[0] = 0$$

$$n=1 \quad y[1] = 0.5y[0] + 4x[-1] \rightarrow y[1] = 0$$

$$n=2 \quad y[2] = 0.5y[1] + 4x[0] \rightarrow y[2] = 4$$

$$n=3 \quad y[3] = 0.5y[2] + 4x[1] \rightarrow y[3] = 2$$

$$n=4 \quad y[4] = 0.5y[3] + 4x[2] = y[4] = 1$$

$$y[n] = 0 \quad n < 2$$

$$y[n] = y[2](0.5)^{n-2} \quad n \geq 2$$

9. The responses of a linear time (shift)-invariant system to specified inputs are defined as follows:

input		reponse
name	symbol	
unit sample	$\delta[n]$	$h[n]$
unit step	$u[n]$	$s[n]$

calculate $h[n]$ for the system, given that
 $s[n] = u[n] - u[n - 5]$.

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

$$u[n-5] = \begin{cases} 1 & n \geq 5 \\ 0 & n < 5 \end{cases}$$

$$s[n] = u[n] - u[n-5]$$

$$s[n] = \begin{cases} 1 & 0 \leq n < 5 \\ 0 & \text{diğer durumlarda} \end{cases}$$

$$h[n] = s[n] - s[n-1]$$

$$s[n] = u[n] - u[n-5]$$

$$s[n-1] = \begin{cases} 0 & n=0 \\ 1 & 1 \leq n < 5 \end{cases}$$

$$h[0] = 1$$

$$1 \leq n < 5 \rightarrow h[n] = 0 \rightarrow 0$$

$$h[n] = s[n] - s[n-5]$$

$$n=5 \rightarrow h[5] = -1$$

10. Find the Fourier transform of $x(t)$.

$$x(t) = \begin{cases} \frac{1}{2}, & -T < t < T \\ 0, & \text{other} \end{cases}$$

Fourier transform

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j2\pi f t} dt$$

$$\rightarrow \int_{-T}^T \frac{1}{2} e^{-j2\pi f t} dt$$

$$\frac{1}{2} \int_{-T}^T e^{-j2\pi f t} dt$$

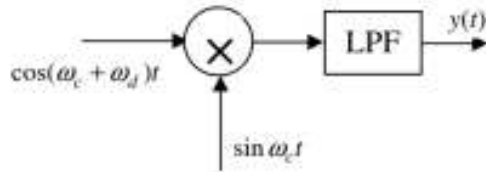
$$X(f) = \frac{1}{2} \left[\frac{e^{-j2\pi f t}}{-j2\pi f} \right]_{-T}^T$$

$$\rightarrow X(f) = \frac{1}{2} \left(\frac{e^{-j2\pi f T} - e^{j2\pi f T}}{-j2\pi f} \right)$$

$\rightarrow \cos(\omega) - j \sin(\omega)$
 $- (\cos(\omega) + j \sin(\omega))$

$$\rightarrow \frac{1}{2} \left(\frac{-2j \sin(2\pi f T)}{-j2\pi f} \right) = \frac{\sin(2\pi f T)}{\pi f}$$

11. Given that the cut-off frequency of the low pass filter (LPF) is ω_c evaluate the output $y(t)$.
 Note: LPF allows frequency values between $-\omega_c < \omega < \omega_c$.



$$x[n] = 2\delta[n+2] - \delta[n+1] + 3\delta[n] - \delta[n-1] + 2\delta[n-2]$$

$X[n]$ 'in fourier dönüşümü

$$X(e^{j\omega}) = 2e^{j2\omega} - e^{j\omega} + 3 - e^{-j\omega} + 2e^{-j2\omega}$$

12. Given that $x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$ is a DTFT pair, evaluate $X(e^{j\omega})|_{\omega=\pi}$ without explicitly finding $X(e^{j\omega})$.

$$x[n] = 2\delta[n+2] - \delta[n+1] + 3\delta[n] - \delta[n-1] + 2\delta[n-2]$$

$$2\delta[n+2] \longrightarrow 2e^{-j\omega 2}$$

$$-\delta[n+1] \longrightarrow -e^{-j\omega 1}$$

$$3\delta[n] \longrightarrow 3$$

$$-\delta[n-1] \longrightarrow -e^{j\omega 1}$$

$$2\delta[n-2] \longrightarrow 2e^{j\omega 2}$$

$$X(e^{j\omega}) \longrightarrow 2e^{-j\omega 2} - e^{-j\omega 1} + 3 - e^{j\omega 1} + 2e^{j\omega 2}$$

$$e^{-j\pi 2} \longrightarrow 1$$

$$e^{-j\pi 1} \longrightarrow -1$$

$$e^{j\pi 1} \longrightarrow -1$$

$$e^{j\pi 2} \longrightarrow 1$$

$$\hookrightarrow X(e^{j\omega}) \longrightarrow 2 + 1 + 3 + 1 + 2 \longrightarrow 9$$

$$\longrightarrow 9 //$$

13. Given that $x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$ and $y[n] \xleftrightarrow{DTFT} Y(e^{j\omega})$ are DTFT pairs, prove the convolution theorem.

$$x[n] * y[n] \xleftrightarrow{DTFT} X(e^{j\omega})Y(e^{j\omega})$$

Konvolüsyon tanımı

$$\sum_{k=-\infty}^{\infty} x[k]y[n-k]$$

DTFT Dönüşümü

$$\sum_{n=-\infty}^{\infty} z[n]e^{-j\omega n}$$

yerine koyma

$$\sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k]y[n-k] \right) e^{-j\omega n}$$

$$\sum_{k=-\infty}^{\infty} x[k] \sum_{n=-\infty}^{\infty} y[n-k] e^{-j\omega n}$$

$n' = n - k$

$$\sum_{k=-\infty}^{\infty} x[k] \sum_{n'=-\infty}^{\infty} y[n'] e^{-j\omega(n'+k)}$$

$$\sum_{k=-\infty}^{\infty} x[k] e^{-j\omega k} \sum_{n'=-\infty}^{\infty} y[n'] e^{-j\omega n'}$$

$$\underbrace{X(e^{j\omega})}_{k=-\infty} \quad \underbrace{Y(e^{j\omega})}_{n=-\infty}$$

14. Find the inverse DTFT of $X(e^{j\omega})$.

$$X(e^{j\omega}) = \cos^2(\omega)$$

$$\cos^2(\omega) \rightarrow \frac{1 + \cos(2\omega)}{2}$$

$$\cos(\omega t) = \frac{1}{2} (e^{j\omega t} + e^{-j\omega t})$$

$$X(e^{j\omega}) \rightarrow \frac{1}{2} + \frac{1}{2} \cos(2\omega)$$

$$\frac{1}{2} \int_{-\pi}^{\pi} \cos(2\omega) \cdot e^{j\omega n} d\omega = \left(\frac{1}{2} (\int_{-\pi}^{\pi} \cos(2\omega) e^{j\omega n} d\omega) \right) \frac{1}{2}$$

$$X[n] \rightarrow \frac{1}{2} \delta[n] + \frac{1}{4} (\delta[n-2] + \delta[n+2])$$

$$X[n] \rightarrow \frac{1}{2} \delta[n] + \frac{1}{4} \delta[n-2] + \frac{1}{4} \delta[n+2]$$

15. Given the 6-point sequence $x[n] = [4, -1, 4, -1, 4, -1]$, determine its 6-point DFT sequence $X[k]$.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn} \quad N=6 \text{ by formula}$$

$$X[k] = k=0,1,2,3,4,5$$

$$k=0$$

$$X[0] = \sum_{n=0}^5 x[n] e^{-j \frac{2\pi}{6} 0 \cdot n} = \sum_{n=0}^5 x[n] = 4 + (-1) + 4 + (-1) + 4 + (-1) = 9$$

$$k=1$$

$$X[1] = \sum_{n=0}^5 x[n] e^{-j \frac{2\pi}{6} 1 \cdot n} = 4 + (-1) e^{-j \frac{2\pi}{6} \cdot 1} + 4 e^{-j \frac{2\pi}{6} \cdot 2} + (-1) e^{-j \frac{2\pi}{6} \cdot 3} + 4 e^{-j \frac{2\pi}{6} \cdot 4} + (-1) e^{-j \frac{2\pi}{6} \cdot 5}$$

$$k=2$$

$$X[2] = 4 + (-1) e^{-j \frac{2\pi}{3}} + 4 e^{-j \frac{4\pi}{3}} + (-1) e^{-j 2\pi} + 4 e^{-j \frac{8\pi}{3}} + (-1) e^{-j \frac{10\pi}{3}}$$

$$k=3$$

$$X[3] = 4 + (-1) e^{-j \pi} + 4 e^{-j 2\pi} + (-1) e^{-j 3\pi} + 4 e^{-j 4\pi} + (-1) e^{-j 5\pi}$$

$$k=4$$

$$X[4] = 4 + (-1) e^{-j \frac{4\pi}{3}} + 4 e^{-j \frac{8\pi}{3}} + (-1) e^{-j 4\pi} + 4 e^{-j \frac{16\pi}{3}} + (-1) e^{-j \frac{20\pi}{3}}$$

$$k=5$$

$$X[5] = 4 + (-1) e^{-j \frac{5\pi}{3}} + 4 e^{-j \frac{10\pi}{3}} + (-1) e^{-j 5\pi} + 4 e^{-j \frac{20\pi}{3}} + (-1) e^{-j \frac{25\pi}{3}}$$

$$X[k] = \begin{cases} 9 \longrightarrow k=0 \\ 8.88 \cdot 10^{-6} \longrightarrow k=1 \\ 0 \longrightarrow k=2 \\ 15 \longrightarrow k=3 \\ 0 \longrightarrow k=4 \\ 8.88 \times 10^{-6} \longrightarrow k=5 \end{cases}$$

16. If the 4-point DFT an unknown length-4 sequence $v[n]$ is $V[k] = [1, 4 + j, -1, 4 - j]$, determine $v[n]$.

$$\underline{\underline{V[n] = \frac{1}{N} \sum_{k=0}^{N-1} V[k] e^{j\frac{2\pi}{N}kn}}}$$

$$N=4$$

$$n=0,1,2,3,4$$

$$V[0]=1$$

$$V[1]=4+j$$

$$V[2]=-1$$

$$V[3]=4-j$$

$$n=0 \rightarrow e^{j0}, e^{j0}, e^{j0}, e^{j0}$$

$$n=1 \rightarrow e^{j0}, e^{j\pi/2}, e^{j\pi}, e^{j3\pi/2}$$

$$n=2 \rightarrow e^{j0}, e^{j\pi}, e^{j2\pi}, e^{j3\pi}$$

$$n=3 \rightarrow e^{j0}, e^{j3\pi/2}, e^{j6\pi/4}, e^{j9\pi/4}$$

$$n=0 \quad \frac{1}{4} (1+4+j-1+4-j) = 2$$

$$n=1 \quad \frac{1}{4} (1-1+1+1) \rightarrow 1/2$$

$$n=2 \quad \frac{1}{4} (1-4-j-1+4+j) \rightarrow -2$$

$$n=3 \quad \frac{1}{4} (1-j-1+j) \rightarrow -\frac{j+1}{4}$$

$$\rightarrow [2, 1/2, -2, -\frac{j+1}{4}]$$

17. Find z-transforms of $x[n]$.

$$x[n] = 6\delta[n] - 7\delta[n-3] - 2\delta[n] - 9\delta[n-5]$$

$$\downarrow$$
$$6$$

$$\downarrow$$
$$7z^{-3}$$

$$\downarrow$$
$$2$$

$$\downarrow$$
$$9z^{-5}$$

$$\longrightarrow 6 - 2 - 7z^{-3} - 9z^{-5}$$

$$\longrightarrow 4 - 7z^{-3} - 9z^{-5}$$

18. If the region of convergence (ROC) for any $x[n] \xleftrightarrow{ZT} X(z)$ z-transform pair includes the unit circle in the complex plane then, $x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$ DTFT pair can also be calculated (converges). Given that the following $X(z)$ includes the unit circle in its region of convergence, evaluate DTFT of $x[n]$ at $\omega = \pi$.

$$X(z) = \frac{z + 2z^{-2} + z^{-3}}{1 - 3z^{-4} + z^{-5}}$$

$$z = e^{j\omega}$$

$$\omega = \pi$$

$$z = e^{j\pi} \rightarrow -1$$

$$\rightarrow \frac{-1 + 2(-1)^{-2} + (-1)^{-3}}{1 - 3(-1)^{-4} + (-1)^{-5}}$$

$$\rightarrow \frac{\emptyset}{-3}$$

$$X(-1) = \emptyset //$$

19. Evaluate $h[n] * x[n]$ using the convolution property of z-transform.

$$h[n] = (0.5)^n u[n]$$

$$x[n] = 3^n u[-n]$$

$$h[n] = (0.5)^n u[n]$$

$$H(z) = \frac{1}{1 - 0.5z^{-1}}$$

$$x[n] = 3^n u[-n]$$

$$X(z) = \frac{1}{1 - 3z}$$

$$y[n] = h[n] * x[n]$$

$$Y(z) = H(z) \cdot X(z)$$

$$Y(z) = \left(\frac{1}{1 - 0.5z^{-1}} \right) \left(\frac{1}{1 - 3z} \right)$$

20. Given that $x[n] \xLeftrightarrow{ZT} X(z)$ is a z-transform pair find $x[n]$.

$$X(z) = 2 + 5(z^2 + z^{-2})$$

$$2 \longrightarrow 2\delta[n]$$

$$5z^2 \longrightarrow 5\delta[n-2]$$

$$5z^{-2} \longrightarrow 5\delta[n+2]$$

$$x[n] = 2\delta[n] + 5\delta[n-2] + 5\delta[n+2]$$

21. Given that $x[n] \xleftrightarrow{ZT} X(z)$ is a z-transform pair
find $x[n]$ for $|z| > 2$.

$$X(z) = \frac{(1)z^2}{(1+3z^{-1}+2z^{-2})z^2}$$

$$X(z) = \frac{z^2}{z^2+3z+2}$$

$\nearrow z+1$
 $\searrow z+2$

$$\frac{z^2}{(z+1)(z+2)} = \frac{A}{z+1} + \frac{B}{z+2}$$

$$z^2 = A(z+2) + B(z+1)$$

$$z^2 = (A+B)z + (2A+B)$$

$$A+B=0$$

$$2A+B=1$$

↓

$$A=1$$

$$B=-1$$

$$\rightarrow \frac{1}{z+1} - \frac{1}{z+2}$$

$$\frac{1}{z+1} = (-1)^n u[n]$$

$$\frac{1}{z+2} = (-2)^n u[n]$$

$$\frac{1}{z+2} = (-2)^n u[n]$$

$$\rightarrow (-1)^n - (-2)^n$$

22. A continuous-time sinusoid $a_1(t) = \cos(\omega_1 t + 0.1\pi)$ is sampled at $f_{s1} = 40 \text{ Hz}$ to give $a_1[n]$, and a second continuous-time sinusoid $a_2(t) = \cos(\omega_2 t + 0.1\pi)$ is sampled at $f_{s2} = 50 \text{ Hz}$ to give $a_2[n]$. If $\omega_2 = 30\pi \text{ rad/s}$, determine ω_1 so that $a_1[n] = a_2[n]$. Assume there is no aliasing when sampling $a_1(t)$.

$$a_1(t) = \cos(\omega_1 t + 0.1\pi)$$

$$f_{s1} = 40 \text{ Hz}$$

$$a_2(t) = \cos(\omega_2 t + 0.1\pi)$$

$$f_{s2} = 50 \text{ Hz}$$

$$\omega = 30\pi$$

$$T_{s1} = \frac{1}{f_{s1}} = \frac{1}{40}$$

$$T_{s2} = \frac{1}{f_{s2}} = \frac{1}{50}$$

$$\cos(\omega_1 n T_{s1} + 0.1\pi) = \cos\left(\frac{\omega_1 n}{40} + 0.1\pi\right)$$

$$\cos(\omega_2 n T_{s2} + 0.1\pi) = \cos\left(\frac{\omega_2 n}{50} + 0.1\pi\right)$$

$$\cancel{\frac{\omega_1 n}{40} + 0.1\pi} = \cancel{\frac{\omega_2 n}{50} + 0.1\pi}$$

$$\omega_1 = \frac{40}{50} \omega_2$$

$$\omega_1 = \frac{4}{5} \omega_2$$

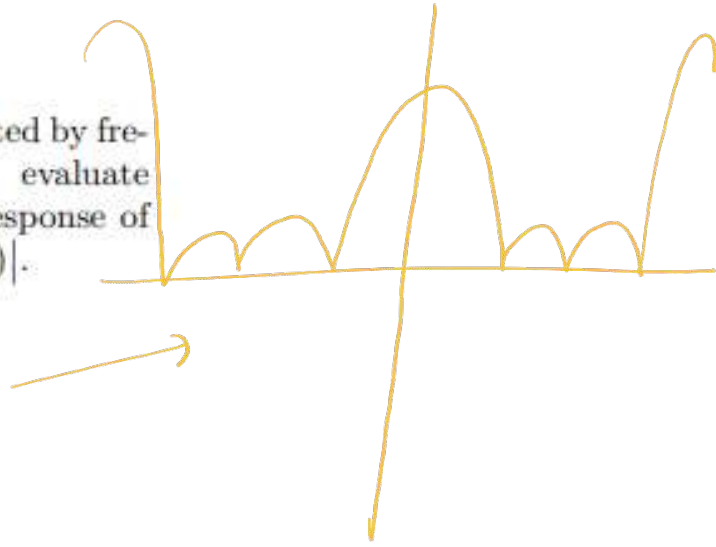
$$\omega_1 = \frac{4}{5} \cdot 30\pi \rightarrow 24\pi //$$

23. A complex bandpass filter is constructed by frequency shifting a running sum filter. evaluate and plot the magnitude frequency response of the complex bandpass filter $|H_B(e^{j\omega})|$.

$$h[n] = \sum_{k=0}^4 \delta[n-k]$$

$$h_B[n] = h[n]e^{j\omega_0 n}$$

$$h_B[n] \xleftrightarrow{DTFT} H_B(e^{j\omega})$$



Bu soruda karmaşık bant geçiren filtrenin frekans kaydırması ile oluşturulmuş bir toplama filtresi üzerinden değerlendirilmesi ve frekans tepkisinin büyüklüğünün çizilmesi isteniyor

$$h[n] = \sum_{k=0}^4 \delta[n-k]$$

frekans kaydırılmış filtre
 $h_B[n] = h[n]e^{j\omega_0 n}$

DTFT
 $h_B[n] \leftrightarrow H_B(e^{j\omega})$

Adım 1: $h[n]$ dizisini bulma

$$h[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \delta[n-4]$$

↓
 5 örnekleme noktasında 1 değerini alır
 diğer noktalarda 0 değerini alır

Adım 2: frekans kaydırılmış filtre

$$h_B[n] = h[n]e^{j\omega_0 n}$$

$$h_B[n] = (\delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \delta[n-4])e^{j\omega_0 n}$$

Adım 3: DTFT hesaplama

$$H_B(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h_B[n]e^{-j\omega n}$$

$$(\delta[n]e^{j\omega_0 n} + \delta[n-1]e^{j\omega_0(n-1)} + \delta[n-2]e^{j\omega_0(n-2)} + \delta[n-3]e^{j\omega_0(n-3)} + \delta[n-4]e^{j\omega_0(n-4)})e^{j\omega n}$$

Adım 4: her term için ayrı ayrı DTFT hesabı

$$e^{j\omega_0 0}e^{-j\omega 0} + e^{j\omega_0 1}e^{-j\omega 1} + e^{j\omega_0 2}e^{-j\omega 2} + e^{j\omega_0 3}e^{-j\omega 3} + e^{j\omega_0 4}e^{-j\omega 4}$$

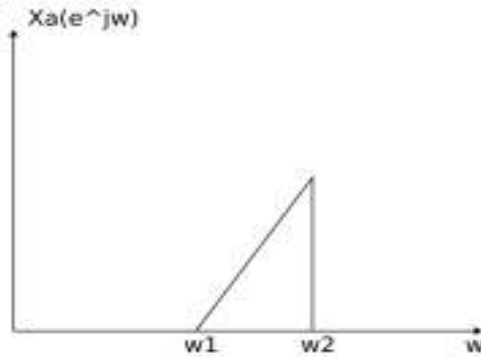
Adım 5: geometrik seri oluşturulur

$$H_B(e^{j\omega}) = \frac{1 - e^{j5(\omega_0 - \omega)}}{1 - e^{j(\omega_0 - \omega)}}$$

$$|H_B(e^{j\omega})| = \left| \frac{\sin(5(\omega_0 - \omega)/2)}{\sin((\omega_0 - \omega)/2)} \right|$$

$$\frac{\sin \delta x}{\delta m x}$$

24. A complex bandpass analog signal $x_a(t)$ has Fourier transform that is non-zero over the range of $[w_1, w_2]$. The signal is sampled to produce the sequence $x[n] = x_a(nT_s)$. What is the smallest sampling frequency that can be used so that $x_a(t)$ may be recovered from its samples $x[n]$.



Karmasık bant geçiren analog sinyalin $X_a(t)$ fourier dönüşümünün belirli bir aralıkta sıfırdan farklı olduğu ve bu sinyalin örneklendirilerek $x[n] = X_a(nT_s)$ dizisine dönüştürüldüğü verilmiştir. Olanca $X_a(t)$ sinyalinin örneklerinden geri kazanabilmesi için kullanılacak en küçük örnekleme frekansını bulmaktır

fourier dönüşümü w_1 ve w_2 arasında sıfırdan farklıdır

$$w_1 \leq w \leq w_2$$

$$B = w_2 - w_1$$

Minimum örnekleme frekansı

$$f_s \geq 2B \rightarrow 2(w_2 - w_1)$$



Bu frekans altında örnekleme yapılması durumunda $X_a(t)$ sinyalinin tüm bilgileri kaybolabilir ve sinyal geri kazanılamaz

25. Plot STFT (Short Time Fourier Transform) representation of a 1D chirp signal with different window sizes. Compare the results with FT of the same signal. Comment on what might be an optimal window size.

Pencere Boyutu etkisi

Küçük pencere boyutları yüksek zaman çözünürlüğü sağlar ancak frekans çözünürlüğü düşer

Büyük pencere boyutları yüksek frekans çözünürlüğü sağlar ancak zaman çözünürlüğü düşer

Optimal pencere boyutu

optimal pencere boyutu, analiz edilen sinyalin özelliklerine ve istenen çözünürlüğe bağlıdır.

Chirp sinyalleri gibi frekansın zamanla değiştiği sinyallerde, orta büyüklükte bir pencere boyutu genellikle iyi bir denge sağlar

Optimal pencere boyutunun belirlenmesi sinyalin doğasına ve analiz gereksinimlerine bağlı olarak değişir

